Angle Chasing

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1 Examples

IMO 2018 P1

Example 1: Let Γ be the circumcircle of acute triangle ABC. Points D and E are on segments AB and AC respectively such that $AD = AE$. The perpendicular bisectors of BD and CE intersect minor arcs AB and AC of Γ at points F and G respectively. Prove that lines DE and FG are either parallel or they are the same line.

Proof: Extend FD, GE to X, Y on Γ.

$\angle AXF = \angle ABF = \angle FDB = \angle ADX$ so $AD = AX = AY$.

So, Y, D, E, X are concyclic with circumcenter A.

Now, $\angle YED = \angle YXD = \angle YXF = \angle YGF$.

Corresponding angles are equal, so $DE \parallel FG\square$.

AoPS solution: me

Indian TST 2019 Practice Test 2 P1

Example 2: Let the points O and H be the circumcenter and orthocenter of an acute angled triangle ABC. Let D be the midpoint of BC. Let E be the point on the angle bisector of $\angle BAC$ such that $AE \perp HE$. Let F be the point such that $AEHF$ is a rectangle. Prove that D, E, F are collinear.

Proof: Let XYZ be the orthic triangle, $AF \cap YZ = U$, and $EF \cap YZ = V$.

Notice that A, H, E, F, Y, Z are concyclic with centre U, and B, C, Y, Z are concyclic with centre D. Additionally,

$$\angle ZVE = 180^\circ - \angle YZE - \angle FEZ = 180^\circ - \angle ZAE - \angle FAZ = 90^\circ$$

Thus, D, E, F, and G all lie on the perpendicular bisector of the radical axis YZ, so we're done. $\square$

AoPS solution: me

IMO Shortlist 2020 G3

Example 3: Let $ABCD$ be a convex quadrilateral with $\angle ABC > 90$, $CDA > 90$ and $\angle DAB = \angle BCD$. Denote by E and F the reflections of A in lines BC and CD, respectively. Suppose that the segments AE and AF meet the line BD at K and L, respectively. Prove that the circumcircles of triangles BEK and DFL are tangent to each other.

Proof: Let X be the reflection of A over BD.

Notice $\angle DA'L = \angle DAL = \angle DFA$, so A' lies on (DFL).

Similarly, $\angle BA'K = \angle BAK = \angle BEK$, so A' lies on (BEK).

Now, notice $\measuredangle A'LD + \measuredangle A'KE = \measuredangle A'LK + \measuredangle A'KL = \measuredangle KA'L$, so we are done.

AoPS solution: me

APMO 2024 P5

Example 4: Line ℓ intersects sides BC and AD of cyclic quadrilateral $ABCD$ in its interior points R and S, respectively, and intersects ray DC beyond point C at Q, and ray BA beyond point A at P. Circumcircles of the triangles QCR and QDS intersect at $N \neq Q$, while circumcircles of the triangles PAS and PBR intersect at $M \neq P$. Let lines MP and NQ meet at point X, lines AB and CD meet at point K and lines BC and AD meet at point L. Prove that point X lies on line KL.

Proof: Notice that M is the Miquel point of complete quadrilateral $ABRSPL$, and N of $CDSRQL$.

So, M, N, L, R, S are concyclic.

$$\begin{aligned}\measuredangle PMN &= \measuredangle PMS + \measuredangle SMN \\ &= \measuredangle PAS + \measuredangle SLN \\ &= \measuredangle BAS + \measuredangle SRN \\ &= \measuredangle RCD + \measuredangle QRN \\ &= \measuredangle RNQ + \measuredangle QRN = \measuredangle PQN\end{aligned}$$

Thus, P, M, N, Q are concyclic. Let $PX \cap (MSRN) = Y$, $QX \cap (MSRN) = Z$.

By Reim's theorem $\triangle LYZ$ and $\triangle KPQ$ are homothetic with centre X, so K, X, L are collinear as desired. $\square$

AoPS solution: me

2 Practise Problems

Here are some practise problems, in increasing order of difficulty.

1. **IMO 2007 P4**

   In triangle ABC the bisector of angle BCA intersects the circumcircle again at R, the perpendicular bisector of BC at P, and the perpendicular bisector of AC at Q. The midpoint of BC is K and the midpoint of AC is L. Prove that the triangles RPK and RQL have the same area.

2. **IMO 2010 P4**

   Let P be a point inside the triangle ABC. The lines AP, BP and CP intersect the circumcircle Γ of triangle ABC again at the points K, L and M respectively. The tangent to Γ at C intersects the line AB at S. Suppose that $SC = SP$. Prove that $MK = ML$.

3. **IMO 2015 P3**

   Let ABC be an acute triangle with $AB > AC$. Let Γ be its circumcircle, H its orthocentre, and F the foot of the altitude from A. Let M be the midpoint of BC. Let Q be the point on Γ such that $\angle HQA = 90^\circ$, and let K be the point on Γ such that $\angle HKQ = 90^\circ$. Assume that the points A, B, C, K and Q are all different, and lie on Γ in this order. Prove that the circumcircles of triangles KQH and FKM are tangent to each other.